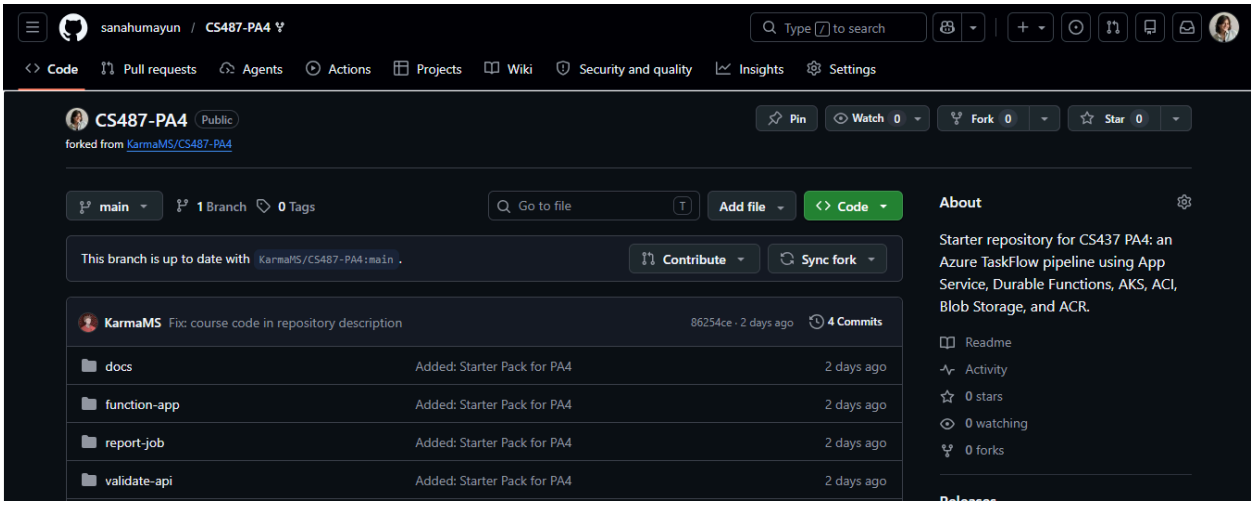
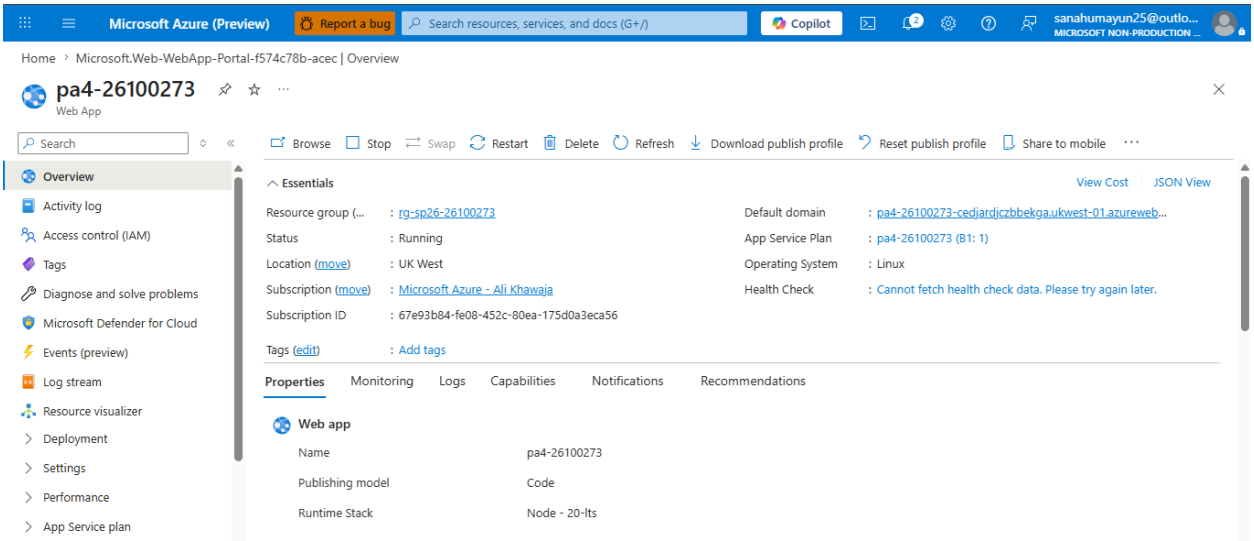
# Task 1: Deploy Web App from GitHub

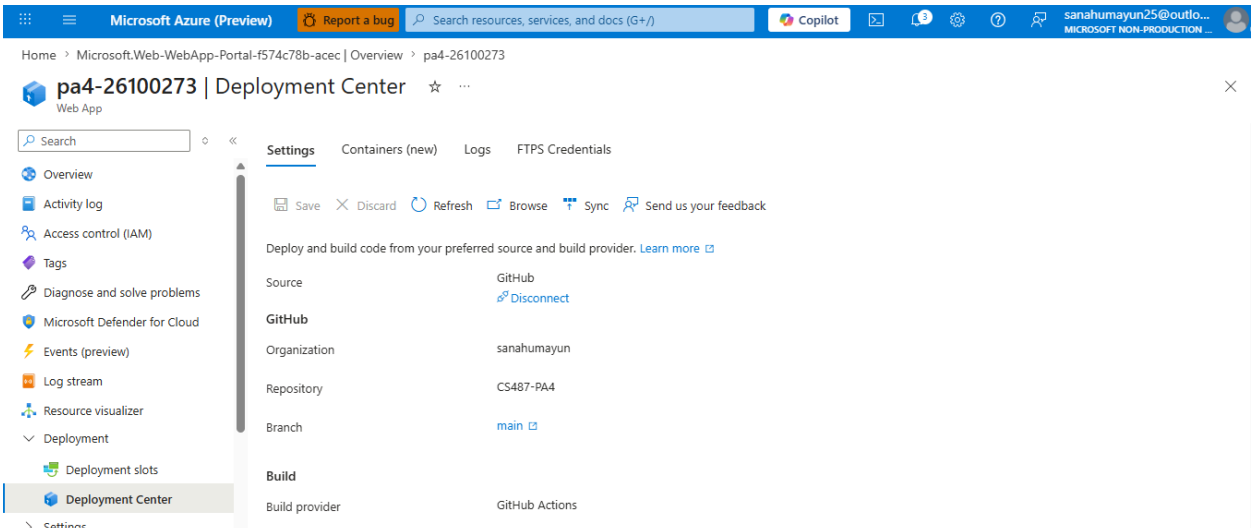
1. Screenshot of your forked GitHub repository



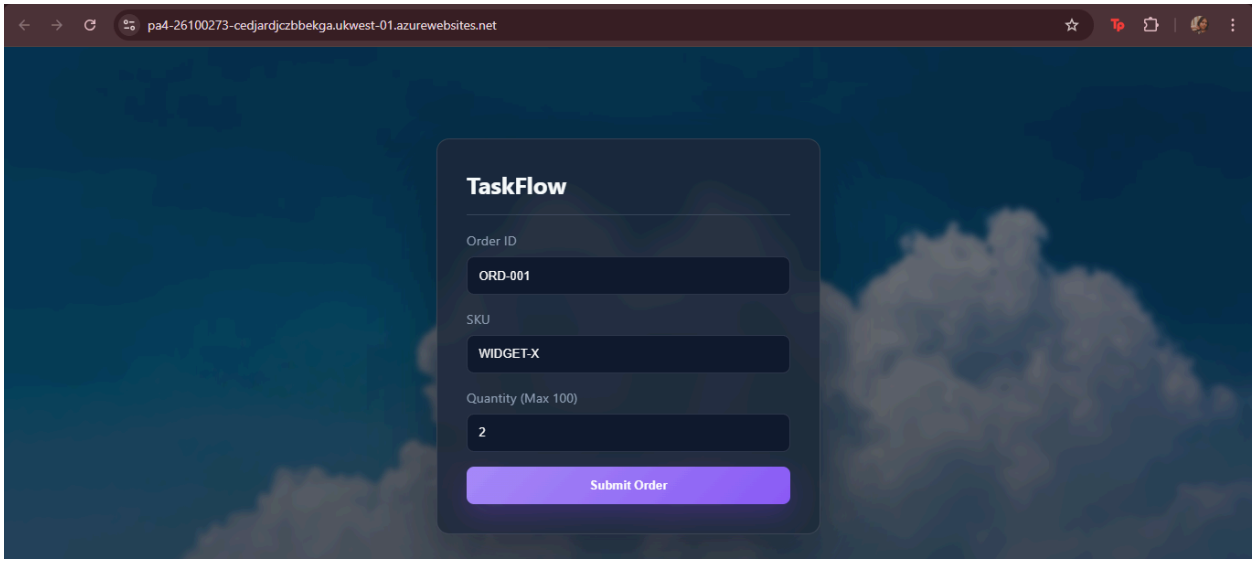

2. Screenshot of Web App overview showing Running status



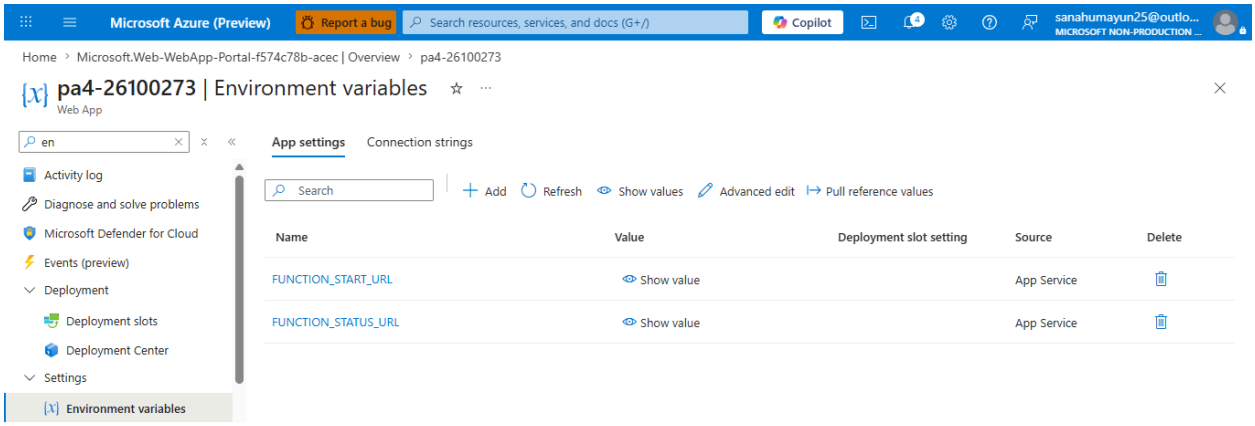

3. Screenshot of Deployment Center showing GitHub configuration



4. Screenshot of the dashboard loading in a browser

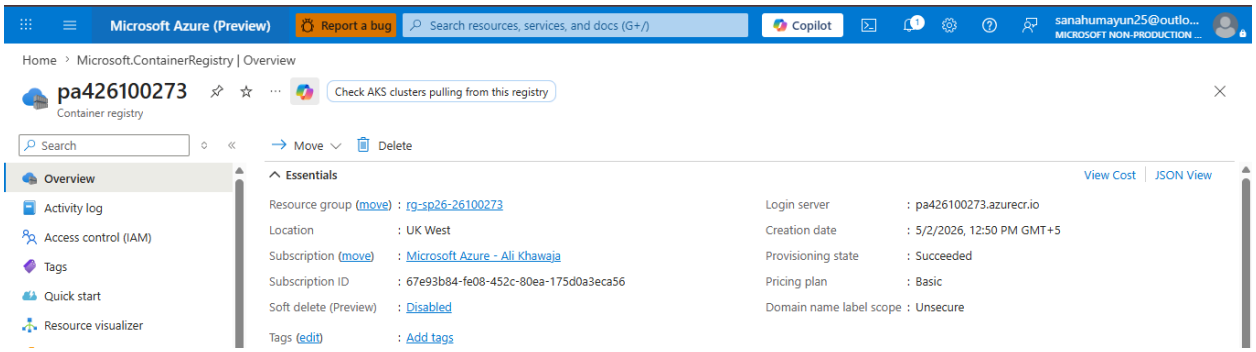



5. Screenshot of Application Settings configured



# Task 2: Create ACR & Push Images

1. Screenshot of ACR overview in the portal showing Succeeded




2. Screenshot of successful Docker builds for all three images

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> docker build -t "pa426100273.azurecr.io/validate-api:v1" ./validate-api
 => => transferring context: 2B                                                                                    0.0s
 => [internal] load build context                                                                                  0.3s
 => => transferring context: 793B                                                                                  0.1s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95757cb45945a99fec8e8084b9c2  16.9s
 => => resolve docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95757cb45945a99fec8e8084b9c2  0.1s
 => => sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d0a363d765d7c799d46035ceb563 249B / 249B                         0.6s
 => => sha256:f3ba2250c52436e41e9f2f3830af92de466fcacc2d848264e1f9fdc76975d803 14.37MB / 14.37MB                   4.8s
 => => sha256:91ff8760033c3b9bc9c3a3bcc6ff66c113ec9e13f94fa2f12ae4b221aae996b9 1.29MB / 1.29MB                     2.7s
 => => sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd 29.78MB / 29.78MB                  10.9s
 => => extracting sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd                          2.9s
 => => extracting sha256:91ff8760033c3b9bc9c3a3bcc6ff66c113ec9e13f94fa2f12ae4b221aae996b9                          0.4s
 => => extracting sha256:f3ba2250c52436e41e9f2f3830af92de466fcacc2d848264e1f9fdc76975d803                          2.0s
 => => extracting sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d0a363d765d7c799d46035ceb563                          0.0s
 => [2/5] WORKDIR /app                                                                                             1.6s
 => [3/5] COPY requirements.txt .                                                                                  0.2s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                      21.2s
 => [5/5] COPY app.py .                                                                                            0.2s
 => exporting to image                                                                                             4.7s
 => => exporting layers                                                                                            2.8s
 => => exporting manifest sha256:a7ea47cc80c16780d839e2e6e99a618fb5f970abdb445040096577be812a8678                  0.0s
 => => exporting config sha256:950ab2c8cae52fad3c885b3bcb80914b8bd95183f5493220186ca503c729d663                    0.0s
 => => exporting attestation manifest sha256:25ae0ef1e288af3b76c69cafe9395f621a4706159ee4accda80fef353491aef7       0.1s
 => => exporting manifest list sha256:3d5dcdeeba0abfa7b83dd1622adbed2d55ee5c4f9252e47c93eec4d5ce5e4938              0.0s
 => => naming to pa426100273.azurecr.io/validate-api:v1                                                            0.0s
 => => unpacking to pa426100273.azurecr.io/validate-api:v1                                                         1.5s
```

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> docker build -t "pa426100273.azurecr.io/report-job:v1" ./report-job
[+] Building 38.3s (10/10) FINISHED                                                                   docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                               0.1s
 => => transferring dockerfile: 284B                                                                               0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                1.3s
 => [internal] load .dockerignore                                                                                  0.1s
 => => transferring context: 2B                                                                                    0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95757cb45945a99fec8e8084b9c2  0.1s
 => => resolve docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95757cb45945a99fec8e8084b9c2  0.1s
 => [internal] load build context                                                                                  0.2s
 => => transferring context: 989B                                                                                  0.1s
 => CACHED [2/5] WORKDIR /app                                                                                      0.0s
 => [3/5] COPY requirements.txt .                                                                                  0.1s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                      26.3s
 => [5/5] COPY generate.py .                                                                                       0.3s
 => exporting to image                                                                                             9.8s
 => => exporting layers                                                                                            7.1s
 => => exporting manifest sha256:8e019664e37846991f4d96df0328c3ecee95db9b2cc45f71004bc0139ba2bebe                  0.1s
 => => exporting config sha256:5badc99f116981386aad5707a2961ca54ee03f35e43e54fc6bfb13587259cf0f                    0.1s
 => => exporting attestation manifest sha256:0e3de2c8a3e26dcefc3e5ab471922433a7f4e9c3c184738a52000a9bf6d17e1d       0.1s
 => => exporting manifest list sha256:061fcb37a08404fa93e2ddaff6e1c1397073ead370e8c83b41aac216d7ca75d5              0.1s
 => => naming to pa426100273.azurecr.io/report-job:v1                                                              0.0s
 => => unpacking to pa426100273.azurecr.io/report-job:v1                                                           2.3s
```

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> docker build -t "pa426100273.azurecr.io/function-app:v1" ./function-app
 => => sha256:8b5c9ed400b721eec77495372ea1d51bccac2cf2d4a7946586d79fc3b1ce606c 110.25MB / 110.25MB                                84.4s
 => => sha256:8715c03abd5b55a9bf49a087f3e225a0e8550c9d79ad233c7a018f207e02a781 14.97MB / 14.97MB                                12.5s
 => => sha256:89320e7119e225692d79aaeb4989a18d7f97daafdb2b3782f84a0a8de31a09de 30.25MB / 30.25MB                                22.2s
 => => extracting sha256:89320e7119e225692d79aaeb4989a18d7f97daafdb2b3782f84a0a8de31a09de                                     7.0s
 => => extracting sha256:8715c03abd5b55a9bf49a087f3e225a0e8550c9d79ad233c7a018f207e02a781                                     1.4s
 => => extracting sha256:8b5c9ed400b721eec77495372ea1d51bccac2cf2d4a7946586d79fc3b1ce606c                                     8.9s
 => => extracting sha256:66f110fb4b1b30f1d6f9b8e4c2ce3f773ca2b2e5db5a0bac9492d4f7e8ca4e48                                    31.6s
 => => extracting sha256:2cff7b2fc77f250a6421dc20adf1efe1122ee84e4d25ba757b215e40f01b53c0                                     0.1s
 => => extracting sha256:cc2ef16d9998ef784bec27db3b0a7d9566c7d6f548209dcbe09dc0540795e303                                     0.1s
 => => extracting sha256:1e291ff5c47c1b3f8bc85bb9531466f57bb4b033b08e838dc7cb966cde82b8d7                                     0.1s
 => => extracting sha256:f6e58f04b6b249aba6c4c5f2e315fb9c81302937ff37447bd3c3a51c34afc91c                                    22.6s
 => => extracting sha256:265a95973510042791267ec78cc4ac86bed9fdda211a9bdcfeb6d24e732b8606                                     3.1s
 => => extracting sha256:3b381515622299dca0079146a171e946b90fa4b705f0b0c1835de0beaaeede92                                    16.9s
 => => extracting sha256:af5ac245ac18c59f1c895aecba7314cba466cc28140f57712597af89190124f0                                    15.7s
 => => extracting sha256:60993470af6a7fe3afdc88644fb0540c417b710ba156a1439e9ae76b6bdd9709                                     2.9s
 => => extracting sha256:83dc37193a8230fb9d5ca60a40c224e76e0b3983c99212a18c7375bde7ce171e                                     0.1s
 => => extracting sha256:48805a8c2422852f23b8914e91fafb3ca5859f10d5b9bcd0b1ade31255ae423c                                     8.9s
 => => extracting sha256:295675d49bd59c704d6862dcb641caeffbe31f6aacbe1dc2f54e4e2c40069353                                     0.1s
 => => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1                                     0.1s
 => => extracting sha256:c2b164019d211e879e19d6bd1a4062f2342a9fcd319758f5bf6863862e5e7486                                     1.6s
 => => extracting sha256:40104569dda32a845c08ff8dc458f60a5358c3fbe93c2e001cc34964b6c71fb8                                     0.1s
 => [internal] load build context                                                                                              0.2s
 => => transferring context: 5.22kB                                                                                            0.1s
 => [2/4] COPY requirements.txt /                                                                                              4.4s
 => [3/4] RUN pip install -r /requirements.txt                                                                                62.6s
 => [4/4] COPY . /home/site/wwwroot                                                                                            0.3s
 => exporting to image                                                                                                         9.8s
 => => exporting layers                                                                                                        5.6s
 => => exporting manifest sha256:93c2a9232a6c36814d3aaae28a022293a546487d58fb0abf5cb232b59b8d4bfb                              0.1s
 => => exporting config sha256:de8da59e22c7ced0ea946de4df6019a7fe8974ace0f41df1eb5397015f4ea075                                0.0s
 => => exporting attestation manifest sha256:50b15e2a94392bfce4be8635609f50fc37de94fe93c7f8100ae68df48b4caae1                  0.1s
 => => exporting manifest list sha256:b5486da3373b93133ecca76ead80cf3a316a36d2e5869aa89e0f21656870ec09                         0.0s
 => => naming to pa426100273.azurecr.io/function-app:v1                                                                        0.0s
 => => unpacking to pa426100273.azurecr.io/function-app:v1                                                                     3.6s
```

3. Screenshot of local test of the validator (POST /validate returning JSON)

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   ESP-IDF                                                    powershell

PS D:\Senior Spring\Cloud Dev\CS487-PA4> docker images
IMAGE                                    ID             DISK USAGE   CONTENT SIZ
E   EXTRA
pa426100273.azurecr.io/function-app:v1   b5486da3373b       2.8GB          748M
B
pa426100273.azurecr.io/report-job:v1     061fcb37a084       282MB         68.7M
B
pa426100273.azurecr.io/validate-api:v1   3d5dcdeeba0a       223MB         54.2M
B
PS D:\Senior Spring\Cloud Dev\CS487-PA4> docker run --rm -p 8080:8080 pa42610027
3.azurecr.io/validate-api:v1
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
INFO:     172.17.0.1:48612 - "POST /validate HTTP/1.1" 200 OK

PS D:\Senior Spring\Cloud Dev\CS487-PA4> Invoke-RestMethod -Method Post -Uri "ht
tp://localhost:8080/validate" -ContentType "application/json" -Body (@{order_id=
"LOCAL-1";items=@(@{sku="ABC";qty=2})} | ConvertTo-Json -Depth 5)

valid reason order_id
----- ------ --------
 True ok     LOCAL-1


PS D:\Senior Spring\Cloud Dev\CS487-PA4>
```

4. Screenshot of successful pushes to ACR for all three images

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> docker push pa426100273.azurecr.io/validate-api:v1
The push refers to repository [pa426100273.azurecr.io/validate-api]
00accd58b9f5: Pushed
0634c5a63e32: Pushed
2632626043e3: Pushed
7ccd73948dde: Pushed
3531af2bc2a9: Pushed
91ff8760033c: Pushed
f3ba2250c524: Pushed
12b7f3d51bad: Pushed
f8ef05ef8cac: Pushed
v1: digest: sha256:3d5dcdeeba0abfa7b83dd1622adbed2d55ee5c4f9252e47c93eec4d5ce5e4938 size: 856
```

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> docker push pa426100273.azurecr.io/function-app:v1
The push refers to repository [pa426100273.azurecr.io/function-app]
4f4fb700ef54: Waiting
af5ac245ac18: Waiting
6394faa04525: Waiting
89320e7119e2: Waiting
cc2ef16d9998: Waiting
60993470af6a: Waiting
af5ac245ac18: Pushed
6394faa04525: Pushed
8aff0b2759af: Pushed
3b3815156222: Pushing  17.83MB/92.57MB
c80d9f07b1c7: Pushed
1e291ff5c47c: Pushed
40104569dda3: Pushed
66f110fb4b1b: Pushing  22.02MB/286.9MB
3b3815156222: Pushed
295675d49bd5: Pushed
f6e58f04b6b2: Pushing  25.17MB/31.53MB
265a95973510: Pushed
66f110fb4b1b: Pushed
48805a8c2422: Pushed
8b5c9ed400b7: Pushing  19.92MB/110.3MB
f6e58f04b6b2: Pushed



8b5c9ed400b7: Pushed

v1: digest: sha256:b5486da3373b93133ecca76ead80cf3a316a36d2e5869aa89e0f21656870ec09 size: 856
```

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> docker push pa426100273.azurecr.io/report-job:v1
The push refers to repository [pa426100273.azurecr.io/report-job]
a5bb850932cd: Pushed
0634c5a63e32: Mounted from validate-api
904ec9676fd2: Pushed
ae49c3a49db6: Pushed
91ff8760033c: Mounted from validate-api
30140cf3bc53: Pushed
f3ba2250c524: Mounted from validate-api
7ccd73948dde: Mounted from validate-api
3531af2bc2a9: Mounted from validate-api
v1: digest: sha256:061fcb37a08404fa93e2ddaff6e1c1397073ead370e8c83b41aac216d7ca75d5 size: 856
```

5. Screenshot of ACR repository list showing validate-api, report-job, and func-app

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> az acr repository list --name pa426100273 --output table
Result
------------
function-app
report-job
validate-api
```

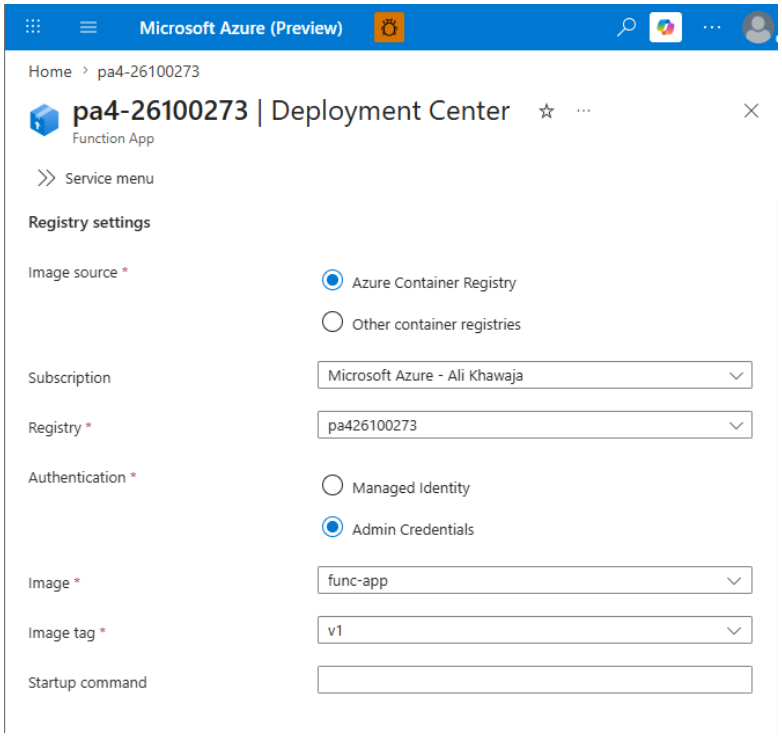
## Task 3: Implement the Durable Orchestrator

1. Screenshot of func start showing all four Durable handlers registered (starter, orchestrator, validate_activity, report_activity)

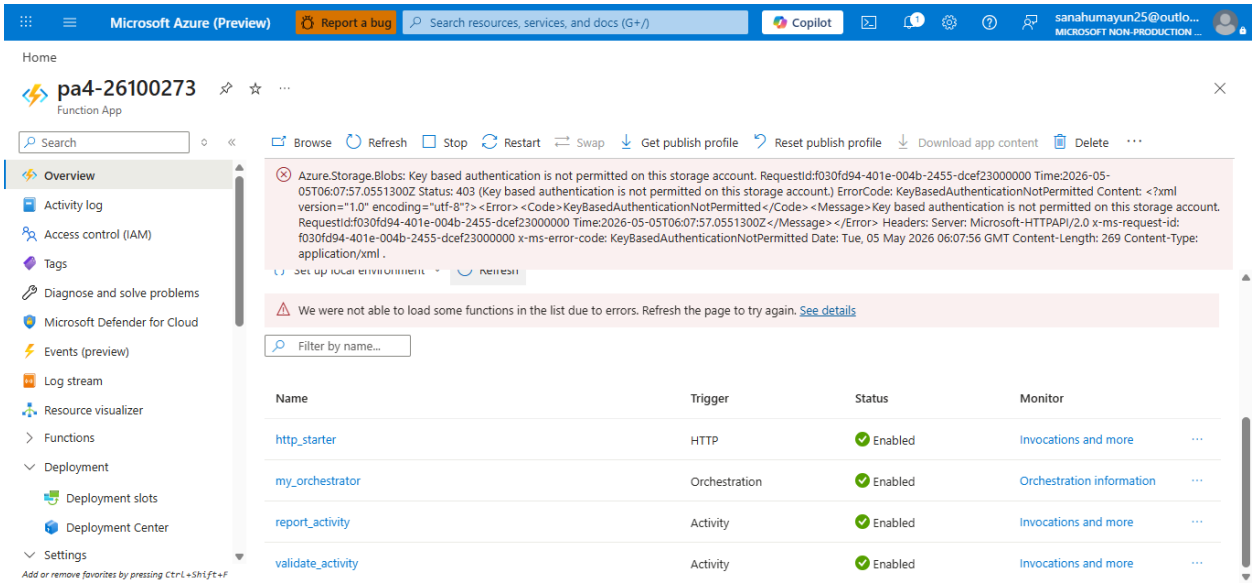
```
PS D:\Senior Spring\Cloud Dev\CS487-PA4\function-app> func start
[2026-05-02T09:00:41.937Z] Worker process started and initialized.

Functions:

        http_starter: [POST] http://localhost:7071/api/orchestrators/my_orchestr
ator

        my_orchestrator: orchestrationTrigger

        report_activity: activityTrigger

        validate_activity: activityTrigger

For detailed output, run func with --verbose flag.
[2026-05-02T09:00:47.068Z] Host lock lease acquired by instance ID '000000000000
00000000000052CBE490'.
```

# Task 4: Deploy Function App as Container

1. Screenshot of Function App in Azure Portal showing the container image configuration



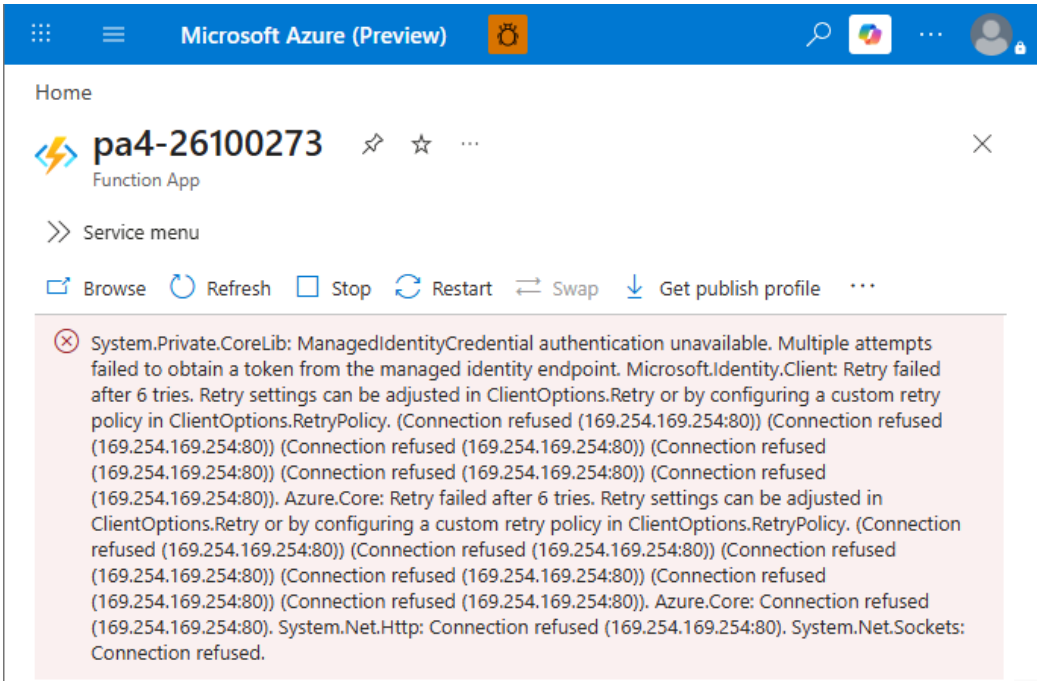

2. Screenshot of the Functions list in the Portal showing http_starter, my_orchestrator, validate_activity, report_activity



3. Screenshot of the curl output showing the orchestration started (instance id and statusQueryGetUri visible)

4. Screenshot of the status query URL JSON showing Failed status at this stage

The Function App container deploys successfully and shows Running status. However, functions fail to load due to `AuthorizationFailure` on `AzureWebJobsStorage`; the subscription's security policies block both connection string access and managed identity access to the auto-generated storage account `rgsp26261002738a9a`. The managed identity `mi-pa4-26100273` does not have pre-granted access to this account. Multiple approaches were attempted including managed identity credentials and direct connection strings.



# Task 5: Azure Kubernetes Service (Validator)

1. Screenshot of kubectl get nodes showing your cluster is running

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> kubectl get nodes
NAME                                STATUS   ROLES    AGE     VERSION
aks-nodepool1-16433285-vmss000001   Ready    <none>   7m18s   v1.34.4
```

2. Screenshot of kubectl get pods showing the validator pod is Running

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> kubectl get pods -w
NAME                                  READY   STATUS                  RESTARTS   AGE
validate-deployment-5b6fd7d56d-tpbtm  1/1     Running                 0          10s
validate-deployment-848b76f8fc-74scq  0/1     Terminating             0          2d9h
validate-deployment-848b76f8fc-74scq  0/1     Terminating             0          2d9h
validate-deployment-848b76f8fc-74scq  0/1     ContainerStatusUnknown  0          2d9h
validate-deployment-848b76f8fc-74scq  0/1     ContainerStatusUnknown  0          2d9h
validate-deployment-848b76f8fc-74scq  0/1     ContainerStatusUnknown  0          2d9h
```

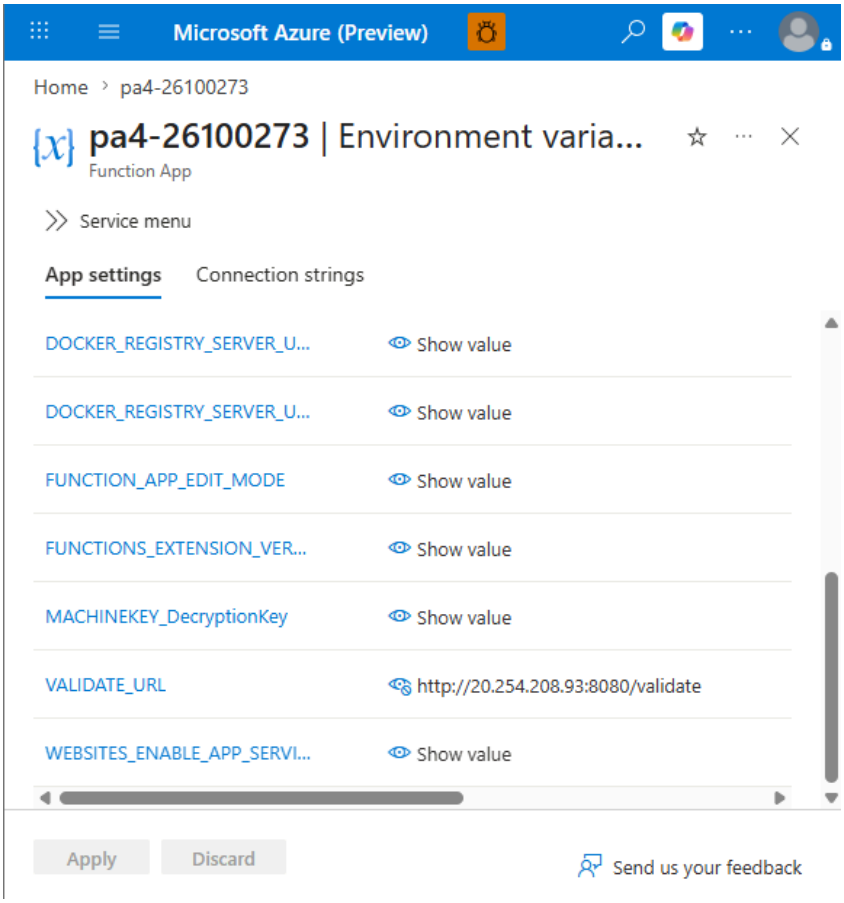
3. Screenshot of kubectl get service showing the assigned EXTERNAL-IP

```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> kubectl get service validate-service
NAME               TYPE           CLUSTER-IP     EXTERNAL-IP     PORT(S)          AGE
validate-service   LoadBalancer   10.0.149.170   20.254.208.93   8080:31549/TCP   2d18h
```

4. Screenshot of curl /health and both curl /validate runs

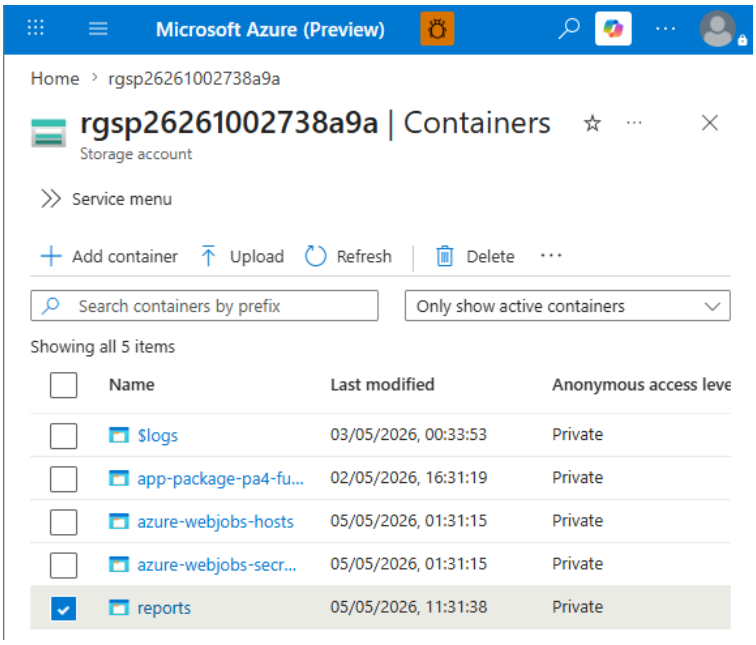
```
PS D:\Senior Spring\Cloud Dev\CS487-PA4> Invoke-RestMethod -Uri "http://20.254.208.93:8080/health" |
 ConvertTo-Json
{
    "status":  "ok"
}
PS D:\Senior Spring\Cloud Dev\CS487-PA4> Invoke-RestMethod -Method POST -Uri "http://20.254.208.93:8
080/validate" -ContentType "application/json" -Body '{"order_id":"O-1001","items":[{"sku":"ABC","qty
":2}]}' | ConvertTo-Json
{
    "valid":  true,
    "reason":  "ok",
    "order_id":  "O-1001"
}
PS D:\Senior Spring\Cloud Dev\CS487-PA4> Invoke-RestMethod -Method POST -Uri "http://20.254.208.93:8
080/validate" -ContentType "application/json" -Body '{"order_id":"O-1002","items":[{"sku":"ABC","qty
":999}]}' | ConvertTo-Json
{
    "valid":  false,
    "reason":  "quantity exceeds limit",
    "order_id":  "O-1002"
}
```

5. Screenshot of Function App Application Setting VALIDATE_URL



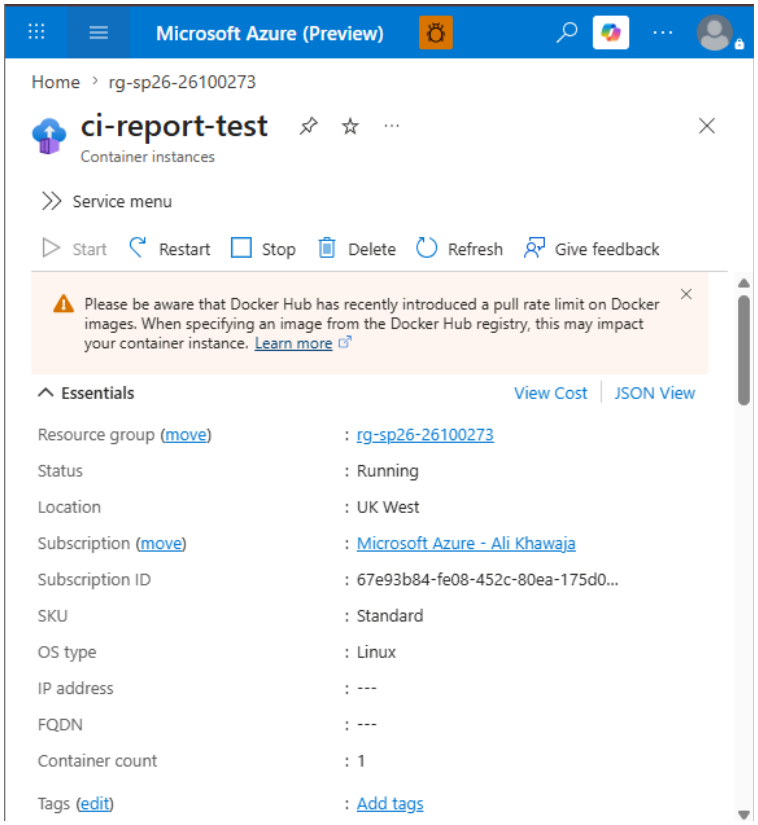
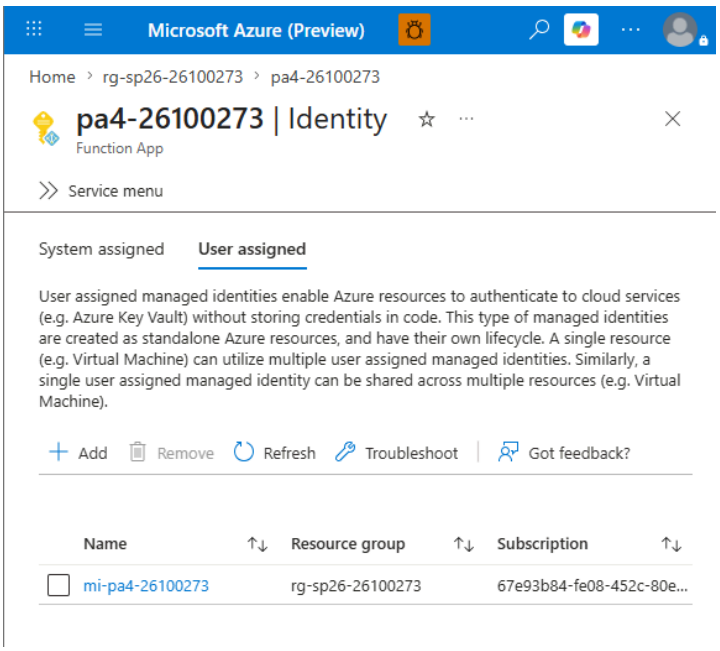
# Task 6: Azure Container Instance

1. Screenshot of the reports blob container after creation




2. Screenshot of az container show output with state: Succeeded after the manual run

```
>> az container logs --resource-group rg-sp26-26100273 --name ci-report-test
    blob.upload_blob(f, overwrite=True)
  File "/usr/local/lib/python3.11/site-packages/azure/core/tracing/decorator.py", line 119, in wrapp
er_use_tracer
    return func(*args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.11/site-packages/azure/storage/blob/_blob_client.py", line 765, in up
load_blob
    return upload_block_blob(**options)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.11/site-packages/azure/storage/blob/_upload_helpers.py", line 195, in
 upload_block_blob
    process_storage_error(error)
  File "/usr/local/lib/python3.11/site-packages/azure/storage/blob/_shared/response_handlers.py", li
ne 184, in process_storage_error
    exec("raise error from None")   # pylint: disable=exec-used # nosec
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "<string>", line 1, in <module>
azure.core.exceptions.HttpResponseError: This request is not authorized to perform this operation.
RequestId:9d672101-001e-004c-7d5b-dc19a6000000
Time:2026-05-05T06:49:37.5757073Z
ErrorCode:AuthorizationFailure
Content: <?xml version="1.0" encoding="utf-8"?><Error><Code>AuthorizationFailure</Code><Message>This
 request is not authorized to perform this operation.
RequestId:9d672101-001e-004c-7d5b-dc19a6000000
Time:2026-05-05T06:49:37.5757073Z</Message></Error>
```

ACI container ran successfully and generated the PDF. Upload failed due to AuthorizationFailure — the managed identity mi-pa4-26100273 was not pre-granted Storage Blob Data Contributor access to the storage account.

3. Screenshot of az container logs showing the report-job's own output (PDF generation lines)
4. Screenshot of the generated PDF listed in the reports blob container
5. Screenshot of the Function App's Identity blade showing the User-assigned identity (mi-pa4-) attached
6. Screenshot of Function App Application Settings showing the REPORT_*, ACR_*, and STORAGE_ACCOUNT_URLvalues (mask the passwords)